# Forecasting PM2.5 over India sixteen hours ahead

A controlled comparison of a Fourier neural operator and a convolutional-recurrent network on WRF-Chem fields, five networks in one fixed pipeline

Why it matters: a chemistry-transport forecast is costly to run, and a learned model that maps recent fields to the next 16 hours would make it cheap

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# Ten hours of fields go in, sixteen hours of PM2.5 come out, for every cell of the map

input

**10 h**

PM2.5 and eight weather fields, 140 x 124 cells



forecast

**16 h**

PM2.5 for every hour and every cell

The guess to beat (persistence)

The last input hour, repeated for 16 hours

Why it is hard

Episodes: rare hours far above the usual daily pattern, about 2% of cell-hours

Scored by

The benchmark score on validation and test; MAE, RMSE and $R^2$ on validation

# Four months of hourly WRF-Chem fields on a 25 km grid over India

ANRF AISEHack 2026, Phase 2, Theme 2 (IIT Delhi)

| Field | Meaning |
|---|---|
| cpm25 | PM2.5 concentration, the target |
| q2, t2 | humidity and temperature at 2 m |
| u10, v10 | wind at 10 m, east and north |
| pblh | boundary layer height |
| psfc | surface pressure |
| swdown, rain | sunlight reaching the ground, rain |

What we have

**140 x 124** cells, 7 to 38 N, 68 to 98 E

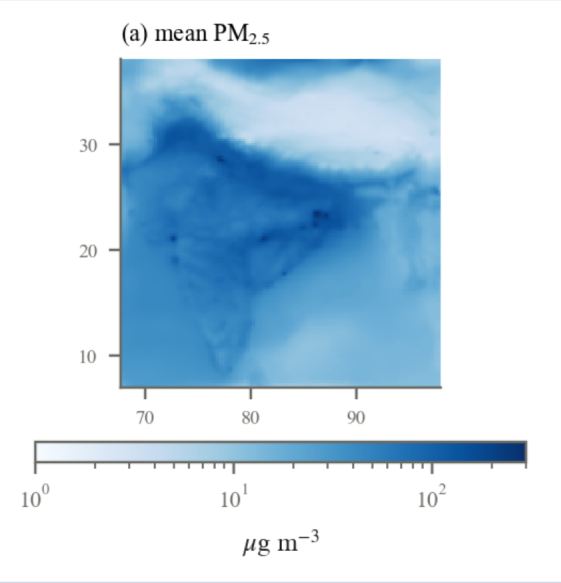
**April, July, October, December 2016**, 715 to 739 hours each

**2,257** training and **214** validation windows

**218** test windows, targets hidden, no timestamps

Simulated by WRF-Chem, not measured at stations

# PM2.5 is skewed, seasonal and slow to change, and each finding shaped the pipeline




Mean PM2.5 on the grid. Hotspots sit over the Indo-Gangetic Plain and the eastern coal belt. The December mean is 2.5 times its median.

Monthly mean, µg/m³

July **15.4**

April **22.4**

October **37.2**

December **59.1**

Skewness 5.1, and 0.13 after a log

So every model works in log(1 + PM2.5)

One hour later PM2.5 is almost the same (autocorrelation 0.978)

So networks predict the change from the last hour

The change grows five-fold by hour 16

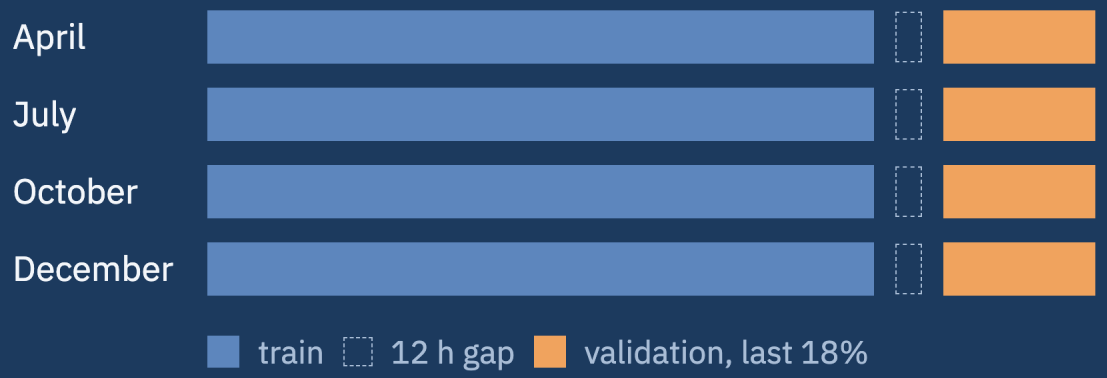
So later hours weigh more in the loss

Wind carries PM2.5 about 10 cells in 16 hours

So an advection channel, and a local view is enough

# A random split leaks: windows one hour apart share 96% of their hours

Each month, split by time




A window is 26 hours long. With a random split, almost every validation window has a training neighbour holding most of its answers.

The last 18% of every month is held out, after a 12-hour gap

Normalization statistics come from the training part only

Test windows have no clock, so the start hour is estimated; validation uses the same estimate

The estimate is within one hour for 97.7% of validation windows

# Every network gets the same 20 channels; physics supplies stagnation and transport

20 inputs per cell and hour

| | | |
|---|---|---|
| **Fields** | PM2.5 and eight weather fields | **9** |
| **Physics** | wind speed, ventilation, advection, PM2.5 tendency | **4** |
| **Daily cycle** | hour as sine and cosine, departure from the usual level at that hour | **3** |
| **Static** | latitude, longitude, PM2.5 climatology, episode threshold | **4** |

Each channel is normalized per cell, with statistics from the training part only.

`ventilation = log(1 + PBLH x wind speed)`

How easily the boundary layer clears. Low ventilation means still air, where pollution builds up.

`advection = -(u d/dx + v d/dy) log(1 + PM2.5)`

How the wind is moving the PM2.5 field right now: the first-order transport term, handed to the network directly.

`tendency = hourly change of log(1 + PM2.5)`

Whether the air is getting dirtier or cleaner.

# Is AFNO V4 still the best network once every network gets the same pipeline?

AFNO V4 was our best model. But while it was built, the pipeline changed too: a leak-free split, official episode masks, physics channels. So its lead mixed two things, the network and the pipeline.

Held fixed for all five networks

- The 20 inputs, the split and the 214 validation windows
- The loss, optimizer, schedule, 16 epochs and seed
- The output: last hour plus a predicted change

The rule, written before any training

**A network is better than AFNO V4 only if it leads by more than 0.003 on validation and also scores higher on the test set.**

A smaller gap is a tie. If validation and test disagree, the result is inconclusive.

Known bias: the recipe was tuned for AFNO V4, so it favours AFNO V4 if anything.

# Five networks, two ideas: mix the whole map at once, or carry the forecast hour by hour

| Network | How it works | Built-in assumption | Weights |
|---|---|---|---|
| ResGRU-UNet | conv U-Net; a ConvGRU steps through the 16 lead hours | local, recurrent | 7.12M |
| AFNO V4 | conv encoder; Fourier (AFNO) blocks mix the whole map | global, spectral | 6.98M |
| FMG-WACLU | ConvLSTM U-Net gated by wind and weather | recurrent, wind physics | 3.45M |
| ConvLSTM | ConvLSTM, a spectral block, a shift along the wind | recurrent, transport | 19.0M |
| patch AFNO | AFNO blocks on 4 x 4 patches | global, spectral | 5.71M |

### Each port is checked against its original

The original network is rebuilt from its notebook; with the same weights, the port gives exactly the same output.

### An untrained network is exactly persistence

The last layer starts at zero, so the output starts as the last hour and training only learns the change.

### One bug fixed in patch AFNO

The original dropped the imaginary half of every filtered spectrum. The port keeps it.

# One recipe for every network, with a loss built from the score itself

The loss, four parts



- **0.40** Huber error on log PM2.5, episode cells count 6 times
- **0.25** relative error (SMAPE) on all cells
- **0.15** relative error on episode cells
- **0.20** one minus the correlation on episode cells

Later lead hours are weighted from 0.7 up to 1.3, because errors grow with lead time.

Against overfitting: 3 to 19 million weights, 2,257 windows

**AdamW**, learning rate 2.5e-4, weight decay 1e-4, one-cycle schedule

**Dropout** on feature maps, 0.1

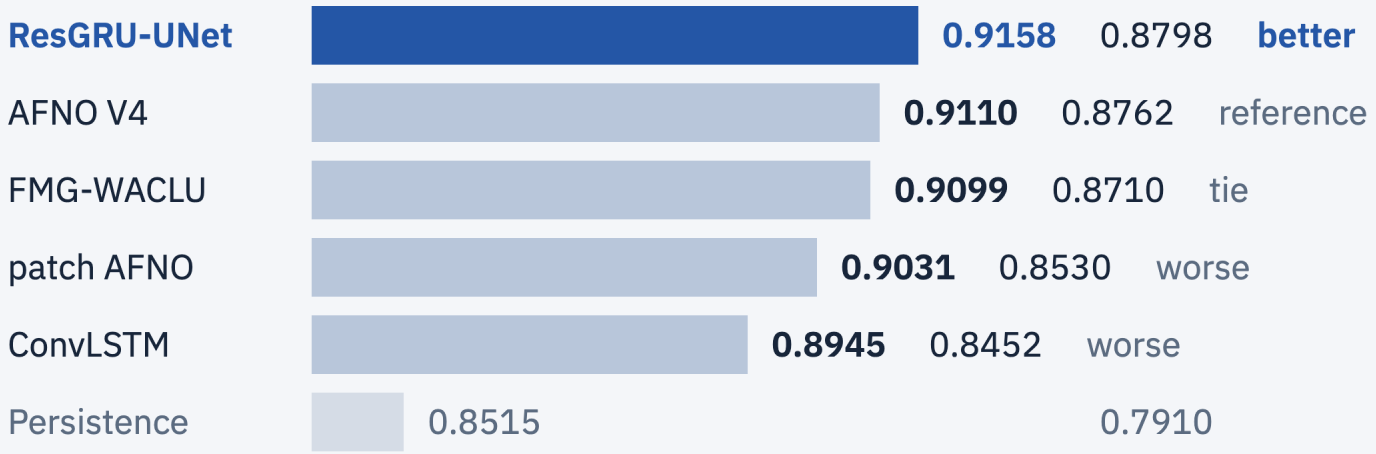
**Averaged weights** (EMA 0.999) are the ones evaluated

**Best epoch** of 16 by validation score, which acts as early stopping

**Gradient clipping** at 1.0; mixed precision on one T4 GPU

# In the same pipeline, ResGRU-UNet is the only network that beats AFNO V4

Benchmark score, seed 42, no calibration. Bars start at 0.84




Bar and first number: validation. Second number: held-out test.

Validation and test rank the five networks in the same order.

The largest network, ConvLSTM with 19 million weights, is last. Size does not decide it.

Check: the AFNO V4 checkpoint, scored again inside the study code, gives 0.91103, exactly its logged value.

# Trained five times each, ResGRU-UNet wins all 25 pairings against AFNO V4

Seed pairs won

**25 of 25**

On validation and on test. Its worst seed (0.9152) is above AFNO V4’s best (0.9123).

Gap of the 5-seed averages

**+0.0048**

On validation. 95% interval 0.0038 to 0.0058, above the 0.003 rule.

Held-out test score

**0.8914**

5 seeds plus calibration. AFNO V4, same setting: 0.8848.

A seed changes only the random starting weights and the order of batches. One run could be luck; five runs with no overlap is not. The interval comes from a block bootstrap: validation windows are resampled in blocks of 13, within each month, 2,000 times.

# ResGRU-UNet cuts RMSE by 31.8% against “no change”; a linear model manages 18.8%

Validation, 214 windows, µg/m³

| Model | MAE | RMSE | R² |
|---|---|---|---|
| Persistence (“no change”) | 10.44 | 26.46 | 0.767 |
| Seasonal naive (24 h earlier) | 11.53 | 26.80 | 0.761 |
| Ridge regression (classical ML) | 8.51 | 21.48 | 0.847 |
| AFNO V4, 5-seed average | 6.99 | 19.45 | 0.874 |
| ResGRU-UNet, 5-seed average | 6.32 | 18.04 | 0.892 |

Test answers are hidden, so MAE, RMSE and R² exist on validation only.

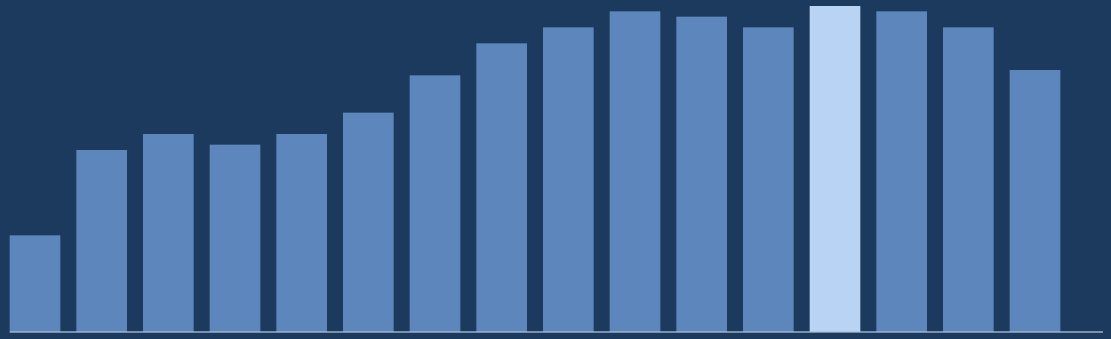
Yesterday’s value is a worse guess than the last hour: seasonal naive loses to persistence.

Ridge uses the same 20 inputs with no network. The deep networks add 13 points of RMSE skill on top of it.

The three baselines, recomputed on a laptop from raw data, match the Kaggle runs exactly.

# The lead holds at every forecast hour, in every month and in 95% of grid cells

Score gap over AFNO V4, by lead hour



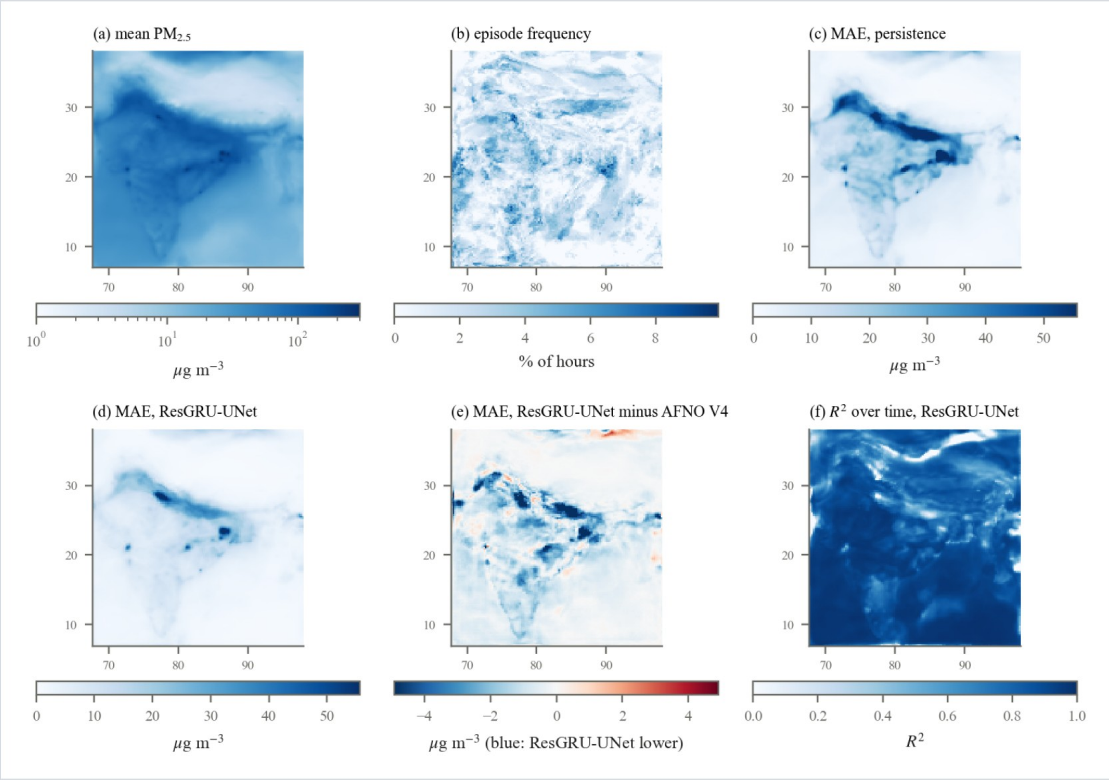
hour 1: +0.0018 hour 13: +0.0061 hour 16

The gap is smallest right after the last observation and grows as the forecast runs further ahead.

| | |
|---|---|
| Lower relative error (SMAPE) | 94.7% of cells |
| Lower MAE | 90.5% of cells |
| Months won | 4 of 4 |
| Typical $R^2$ in a cell | 0.872 against 0.846 |

5-seed averages over the 214 validation windows. Persistence has a typical per-cell $R^2$ of 0.694.

# Errors pile up over the Indo-Gangetic Plain, and that is where ResGRU-UNet gains most




Averages over 214 validation windows, 5 seeds

**(a) and (c): pollution and "no change" error**

Both are highest along the plain below the Himalaya and over the eastern coal belt.

**(d): ResGRU-UNet error**

Same colour scale as (c). Most of the error is gone; a few hotspots remain.

**(e): ResGRU-UNet minus AFNO V4**

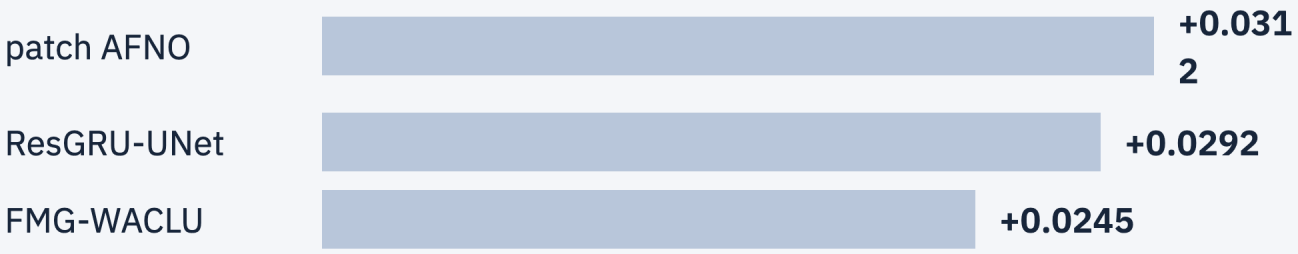
Blue means ResGRU-UNet is more accurate. The map is blue almost everywhere, deepest over the plain.

**(f): R² through time, per cell**

High over land; weakest over the mountains.

# Fixing the pipeline gained 6 to 8 times more than swapping the network

Gain in test score, same network, its own pipeline to the shared one




Gain from swapping AFNO V4 for ResGRU-UNet, same pipeline




Seed 42, no calibration. Each pipeline gain bundles every pipeline change.

What the shared pipeline changed

- Validation that cannot leak and copies the test conditions
- Physics channels for stagnation and transport
- A loss built from the score, model picked by the exact score
- Over eight iterations the test score rose from 0.8218 to 0.8914

# Why does it win? Four single changes were tested, and none explains the gap alone

| Guess, written before the run | The one change | Shift in score | Needed |
|---|---|---|---|
| AFNO V4 loses detail in time | its skips keep all 10 input hours | +0.0011 | 0.0020 |
| The FFT wraps the map edges | AFNO blocks run on a padded map | -0.0008 | 0.0020 |
| Thresholding removes detail | soft threshold switched off | -0.0005 | 0.0020 |
| ResGRU-UNet needs its recurrence | ConvGRU replaced by plain conv blocks | -0.0017 | 0.0021 |

### The test for each change

It counts only if it moves validation by more than twice the seed-to-seed spread, and at least 0.002. The gap to explain is 0.0048.

### What we did find, not planned in advance

At the same training loss, AFNO V4's validation loss is 6.8% higher. It fits the training months more closely and carries over less well.

# We let gradient boosting correct the network; tested fairly, it did not beat the network alone

deep learning
ResGRU-UNet, 5-seed forecast

+

machine learning
LightGBM predicts the network's error

LightGBM sees 33 inputs per cell: the 20 channels, recent PM2.5, the network's forecast, how much the 5 seeds disagree, and the lead hour.

### The rule, written before the run

Fit on three months, predict the fourth. Better only if the overall score is higher and it wins at least 3 of 4 months.

Benchmark score on the held-out month

| Month | Network | Hybrid | Change |
|---|---|---|---|
| April | 0.9162 | 0.8999 | -0.0162 |
| July | 0.9091 | 0.8940 | -0.0151 |
| October | 0.9271 | 0.9203 | -0.0068 |
| December | 0.9249 | 0.9061 | -0.0188 |
| All four | 0.9193 | 0.9051 | -0.0142 |

RMSE improved a little (18.04 to 17.75), but the error on episode cells grew from 0.235 to 0.305. Verdict by the rule: no gain, so the hybrid is not used.

# Get the pipeline right first; then a plain recurrent U-Net beats the Fourier network

## What the study does not show

The data is a simulation, so we emulate WRF-Chem, not the real air

Validation hours are the ends of the training months; another year is untested

Each ablation ran on one seed

Test scores were submitted after the competition deadline

## What we take away

**The pipeline** gained 6 to 8 times more than the network.

**ResGRU-UNet** beat AFNO V4 on every seed pair, lead hour, month and regression metric.

**No single part** explains the gap, and a boosted correction did not add to it.

**Every verdict** followed a rule written before the run.

# Who built what

| | |
|---|---|
| **Ishan Goyal** | Data analysis, the 20 input channels, normalization, the hour estimate, the leak-free split and episode masks |
| **Vetriselvan R** | Study design and decision rules; AFNO V4, ResGRU-UNet, the ports and their check, the ablation variants; the round-one, final-round, ablation and hybrid runs; the report |
| **Hrishi Dongre** | Loss, optimizer and schedule, weight averaging, checkpoints and resume, the notebook builders and the seed runs |
| **Vedant Madne** | Seed ensembles and calibration, the persistence, seasonal-naive and ridge baselines, and MAE, RMSE and $R^2$ |
| **Pampana Sree Adithya** | 5-seed statistics and block bootstrap, the ablation verdict, figures and error maps |